Recommended Assessment

Color Sensor

Analyzing Raw Color Sensor Data

1. When you covered the color sensor with folded white paper, all sensor channels dropped near zero. Why is this step important, and what does it tell you about the sensor?

2. When adjusting individual R, G, and B LEDs, which sensor channels respond most strongly for each LED, which color LED produced the largest response in the clear channel? Refer to the spectral sensitivity of the VEML3328 in your answer.

3. When combining LEDs, what did you observe about the RGB-to-Clear ratios, and what does this tell you about the sensor's behavior?

4. When increasing all three LEDs toward white, the Clear channel saturates first. Why does this happen, and why is identifying the “white reference” important for calibration?

Calibration

1. How did you compute the theoretical XYZ values for the LEDs, and why do these theoretical values differ from real LED XYZ measurements?

2. You may have observed a green-tinted white after applying the calibrated M. Explain why this green bias appears and how calibration procedure contributed to it.?

3. Provide two improvements that would reduce the green bias in future calibrations.

4. Compare the direct sensor to sRGB mapping matrix with the XYZ mapping method. What are the primary advantages and limitations of each?

5. Why does least-squares fitting remain useful and why might the sensor-detected color still differ slightly from the displayed screen color after applying a fitted matrix?

Color Sensor Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	